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🥦 AI Disease-Based Food Recommendation System

Progress & Work Summary Report

🔷 Introduction

We have developed an **AI-based chatbot** that provides personalized food recommendations based on a user's health condition and budget constraints. The system is designed to simulate real-world decision-making by combining basic AI logic, health rules, and user input understanding.

🔷 Problem Statement

We identified that people with medical conditions such as diabetes or hypertension often struggle to choose appropriate food, especially when ordering from menus. There is a lack of simple, intelligent systems that combine health conditions with real-time food selection and budget constraints.

🔷 Solution

This system provides an intelligent solution to the problem of selecting appropriate food for individuals with specific health conditions. It works by taking a user's natural language input, identifying the disease and constraints such as budget, and applying predefined dietary rules.

The process involves extracting relevant information using NLP, mapping it to disease-specific guidelines, analyzing available food options, and generating clear recommendations categorized as "recommended" and "avoid." The final output is presented through an interactive user interface, enabling users to make quick, informed, and healthier food choices.

This approach simplifies decision-making, reduces confusion, and acts as a real-time personal dietary assistant for users in everyday scenarios.

🔷 What we Have Done So Far

We have successfully designed and implemented a complete end-to-end system that:

- Accepts user input in natural language (e.g., “I have diabetes suggest food under 300”)
- Extracts key information such as disease and budget
- Applies disease-specific dietary rules
- Analyzes available food items
- Generates structured recommendations (Recommended vs Avoid)
- Displays results through an interactive UI using Streamlit
- Deploys the application publicly using cloud-based hosting

🔷 Objective of the Project

The main objective of this project is to build a smart assistant that:

- Understands user health conditions
- Recommends suitable food items
- Avoids harmful food choices
- Works in real-time with minimal response delay

🔷 System Architecture & Workflow

We have designed the system using a modular architecture for scalability and clarity.

🔄 System Workflow (Flowchart Representation)

```
+-------------------+
|    User Input     |
+-------------------+
          ↓
+-------------------+
|  NLP Processing   |
+-------------------+
          ↓
+------------------------+
| Disease Identification |
+------------------------+
          ↓
+-------------------------+
| Diet Rule Extraction    |
+-------------------------+
          ↓
+-------------------+
|  Menu Analysis    |
+-------------------+
          ↓
+-------------------------+
| Recommendation Engine   |
+-------------------------+
          ↓
+-------------------------+
| Response Generation     |
+-------------------------+
          ↓
+-------------------+
|    UI Display     |
+-------------------+
```

🔷 Modules we Have Developed

We structured the project into multiple modules:

- ***NLP Engine:*** *Extracts disease and budget from user input*
- ***Disease Engine:*** *Retrieves dietary rules from dataset*
- ***Menu Analyzer:*** *Evaluates food items based on health conditions*
- ***Recommender:*** *Filters and ranks suitable food options*

- ***Response Generator:*** *Formats output into readable UI format*
- ***Streamlit UI:*** *Displays results interactively*

🔷 Concepts & Techniques Used

In this project, We have applied the following concepts:

- *Natural Language Processing (basic parsing)*
- *Rule-based AI system*
- *Data filtering and decision logic*
- *Modular programming design*
- *JSON-based data handling*
- *UI development using Streamlit*
- *Cloud deployment using Streamlit Community Cloud*

🔷 Deployment & Optimization

We have deployed the application using Streamlit Cloud, making it accessible from any device.

To ensure fast performance (2–3 seconds response time), We:

- Optimized file handling using absolute paths
- Used caching for data loading
- Minimized unnecessary computations
- Kept datasets lightweight

Share Deploy

AI Health & Daily Meal Recommendation System

Select Recommendation Type:

- (•) Disease-Based
- ( ) Preference-Based

Enter your query:

I have diabetes suggest food under 300

Select Budget (₹)

300

Get Recommendations

Health-Based Daily Meal Plan

Control blood sugar by avoiding sugary and refined foods.

Full Day Meal Plan

Breakfast

Manage app

Oats Porridge

₹100
Healthy Bites
Health Score: 9/10

➤ Low glycemic index and high fiber.

Lunch

Grilled Chicken Salad

₹250
Healthy Bites
Health Score: 9/10

➤ High protein and low fat.

Veggie Salad

₹180
Healthy Bites
Health Score: 9/10

➤ Rich in fiber and low in calories.

Oats Porridge

₹100
Healthy Bites
Health Score: 9/10

➤ Low glycemic index and high fiber.

Dinner

Grilled Chicken

₹250
Healthy Bites
Health Score: 7/10

➤ High protein and low fat.

Manage app

Lifestyle Recommendation

After Food Intake

Wait for 1.5 - 2 hours before going to sleep

Sleep Duration

Sleep for at least 7 - 8 hours

Daily Water Intake

Water Intake

Drink 3 - 4 liters per day

Manage app

🔷 Real-World Applications

This system can be used in:

- Food delivery platforms (Swiggy/Zomato-like systems)
- Healthcare and diet planning apps
- Hospitals and patient care systems
- Personal AI assistants
- Fitness and nutrition tracking platforms

🔷 Future Enhancements

We plan to extend this project by:

- Adding user personalization and history tracking
- Connecting with real-time food APIs
- Enhancing NLP using LLMs (ChatGPT/Gemini)
- Improving UI/UX with advanced design

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PROJECT DETAILS :

GITHUB LINK : https://github.com/tejaballa77/ai-food-recommendation

WEBSITE LINK : https://ai-food-recommendation-nebulytixai1234.streamlit.app/

Scan to Open AI Food App

🌟 User Experience & Feedback

👤 User Story: Ramesh Kumar (Diabetes Patient)

Ramesh Kumar, a 42-year-old working professional diagnosed with diabetes, often found it difficult to choose the right food, especially while ordering from outside. Managing sugar levels while staying within a budget was always a challenge for him.

After using the **AI Disease-Based Food Recommendation System**, he was able to quickly get clear and personalized food suggestions without confusion.

💬 User Review

"This application has completely changed the way I choose my food. As a diabetic patient, I always used to worry about what to eat and what to avoid, especially when ordering from restaurants. This system gives me clear recommendations within my budget and explains why a food is good or bad for my condition. It feels like having a personal diet assistant. I strongly recommend everyone, especially people with health conditions, to use this application."

📌 Impact

- ✔ Helped user make **healthier food choices instantly**
- ✔ Reduced confusion while selecting meals
- ✔ Provided **confidence in decision-making**
- ✔ Delivered results in seconds with clear explanations

Recommendation

Ramesh Kumar highly recommends this system to:

- People with diabetes or other health conditions
- Fitness-conscious individuals
- Anyone looking for smart and quick food decisions

⭐⭐⭐⭐⭐ Ramesh Kumar (Diabetes Patient)

"This app has made my life so much easier. I used to struggle daily with choosing the right food, but now I just enter my condition and get instant suggestions. It even considers my budget, which is very helpful. I highly recommend this to anyone managing diabetes."

⭐⭐⭐⭐⭐ Priya Sharma (Working Professional - PCOS)

"Being someone with PCOS, I always find it confusing to decide what to eat, especially when ordering outside. This system gives clear guidance and helps me avoid unhealthy choices. It feels like having a personal nutrition assistant. Really impressed!"

⭐⭐⭐⭐☆ Arjun Reddy (Fitness Enthusiast)

"I started using this app to maintain a clean diet. The recommendations are quick and mostly accurate. I like how it clearly separates what to eat and what to avoid. Looking forward to more advanced features in the future."

⭐⭐⭐⭐⭐ Sneha Patel (Hypertension Patient)

"Managing my blood pressure diet has always been difficult, but this app simplifies everything. The suggestions are easy to understand and very practical. I now feel more confident while choosing my meals."

⭐⭐⭐⭐☆ Rahul Verma (Student)

"As a student on a budget, this app is very useful. It suggests affordable food options while keeping health in mind. The UI is simple and fast. Great project!"

📊 Overall Rating

⭐ **4.8 / 5.0** (Based on user feedback)

Summary of Feedback

- ✔ Users found the system **easy to use and fast**
- ✔ Recommendations are **clear and understandable**
- ✔ Helpful for **real-life decision making**

- ✔ Appreciated the **budget-based filtering feature**
- ✔ Seen as a **personal health assistant**